SOLVED PRACTICE WORKBOOK

Economy Topic 31: Solved Practice Workbook

Solved Practice Workbook

LEARNING PATH

CONCEPT
Define precisely

ARGUMENT
Build the case

CRITIQUE
Test the limits

APPLICATION
Write the answer

Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Economy Topic 31: Solved Practice Workbook

Status cutoff: 10 September 2026.

Practice contract: Exactly 32 original MCQs appear before PYQs. Correct answers rotate A -> B -> C -> D eight times. Every option has a question-specific explanation and every MCQ has a unique trap.

BASIC MCQS / REMEDIATION

Q1. Which statement correctly distinguishes primary and secondary energy?

A. Crude oil is primary, while petrol and electricity are secondary carriers. B. Electricity is always primary because it powers final use. C. Hydrogen is a geological primary resource in energy accounts. D. Final energy is measured before conversion losses.

Answer: A

Option explanations:

- **A is correct:** Crude exists before conversion; petrol and electricity are produced carriers.
- **B is incorrect:** Electricity normally results from conversion of a primary source.
- **C is incorrect:** Hydrogen must be produced and therefore acts as a carrier.
- **D is incorrect:** Final energy reaches the user after upstream conversion and network losses.

UPSC trap: Primary, secondary and final describe positions in a conversion chain.

Q2. Which classification is most accurate?

A. All biomass is non-commercial energy. B. Commercial/non-commercial and modern/traditional are separate axes. C. Every renewable source is non-commercial. D. Traditional fuel use proves absence of an electricity connection.

Answer: B

Option explanations:

- **A is incorrect:** Modern pellets and biogas may be traded commercially.
- **B is correct:** A fuel can be commercial yet traditional, or renewable yet modern.
- **C is incorrect:** Solar and wind electricity are commonly sold through markets and contracts.
- **D is incorrect:** Fuel stacking can persist after electrification because of affordability or reliability.

UPSC trap: Do not collapse market status into technology quality.

Q3. At 31 December 2025, what did the 51.93% figure denote?

A. Non-fossil share of total primary energy. B. Renewable share of final energy consumption. C. Non-fossil share of installed electricity capacity. D. Non-fossil share of April-December electricity generation.

Answer: C

Option explanations:

- **A is incorrect:** The Ministry table concerns installed power capacity, not primary energy.
- **B is incorrect:** No final-consumption denominator appears in that capacity table.
- **C is correct:** The annual report labels total non-fossil capacity as 51.93% of installed capacity.
- **D is incorrect:** April-December 2025 non-fossil generation was 30.41%, a separate flow.

UPSC trap: Capacity share and generation share use different denominators and periods.

Q4. A 100 MW battery with 400 MWh usable energy can ideally discharge at rated power for approximately:

A. One hour B. Two hours C. Three hours D. Four hours

Answer: D

Option explanations:

- **A is incorrect:** One hour would use only 100 MWh of the stated energy.
- **B is incorrect:** Two hours would use 200 MWh, half the stated store.
- **C is incorrect:** Three hours would use 300 MWh, below the stated store.
- **D is correct:** Energy divided by power gives four hours before efficiency and operating limits.

UPSC trap: Storage must state both MW and MWh.

Q5. Which statement about PLF is correct?

A. It measures utilisation of thermal capacity over a period, not thermal efficiency. B. It equals coal's share of total generation. C. It is the maximum demand met by a system. D. It measures billing and collection performance.

Answer: A

Option explanations:

- **A is correct:** PLF compares actual thermal output with output possible at rated capacity.
- **B is incorrect:** Generation share is an energy-mix ratio with another denominator.
- **C is incorrect:** Peak demand met is an instantaneous power statistic.
- **D is incorrect:** Billing and collection enter AT&C and revenue metrics.

UPSC trap: Low PLF has several possible causes; do not diagnose efficiency from it.

Q6. Which proposition about deficits is valid?

A. Zero peak deficit proves zero local outages. B. Peak deficit and energy deficit measure different dimensions of shortage. C. Energy deficit is measured only in MW. D. Peak deficit equals the ACS-ARR gap.

Answer: B

Option explanations:

- **A is incorrect:** National peak balance can coexist with local distribution failures.
- **B is correct:** Peak deficit concerns MW at the maximum interval; energy deficit concerns cumulative supply.
- **C is incorrect:** Energy deficit uses energy units or a percentage of requirement.
- **D is incorrect:** ACS-ARR is financial under-recovery in Rs/kWh.

UPSC trap: National adequacy is not identical to local service reliability.

Q7. Why is security-constrained dispatch broader than a simple merit order?

A. It ignores variable cost. B. It applies only to renewable generators. C. It includes congestion, reserves and technical operating limits. D. It fixes retail tariffs.

Answer: C

Option explanations:

- **A is incorrect:** Variable cost remains important within the eligible set.
- **B is incorrect:** Dispatch covers all scheduled resources under applicable rules.
- **C is correct:** Grid security, ramping, minimum load and congestion constrain pure cost ranking.
- **D is incorrect:** Retail tariff is determined by the relevant commission.

UPSC trap: Merit order is nested inside technical and security constraints.

Q8. Which institution-function pair is correct?

A. CEA - retail tariff adjudication B. CERC - household electrification execution C. SERC - national load despatch D. Grid Controller India - national and regional system operation

Answer: D

Option explanations:

- **A is incorrect:** CEA is the technical planning and standards authority.
- **B is incorrect:** CERC regulates specified inter-state and central matters.
- **C is incorrect:** SERC regulates state matters; SLDC performs state system operation.
- **D is correct:** Grid Controller India operates NLDC and regional load-despatch functions.

UPSC trap: Separate technical authority, regulator, network utility and operator.

Q9. What is the Electricity (Amendment) Bill, 2025 status at the cutoff used here?

A. A draft proposal, with no located enactment replacing the 2003 Act. B. A constitutional amendment already in force. C. A CERC tariff regulation. D. A state code automatically binding nationwide.

Answer: A

Option explanations:

- **A is correct:** The Ministry circulated a draft; no enactment was located by 10 September 2026.
- **B is incorrect:** This is not a constitutional-amendment measure.
- **C is incorrect:** A parliamentary Bill is not a commission regulation.
- **D is incorrect:** A state code cannot enact a central parliamentary Bill.

UPSC trap: Proposal, passage, assent and commencement are separate stages.

Q10. Which tariff statement is correct?

A. A low consumer tariff always proves low system cost. B. Explicit subsidy and cross-subsidy have different payers and incidence. C. Regulatory assets eliminate the underlying cost. D. Fixed charges vary only with each kWh.

Answer: B

Option explanations:

- **A is incorrect:** Low tariffs may conceal subsidy, debt or deferred maintenance.
- **B is correct:** Budget subsidy is paid by government; cross-subsidy is borne by another class.
- **C is incorrect:** A regulatory asset defers recovery and usually carries future cost.
- **D is incorrect:** Fixed charges recover costs not proportional to current energy use.

UPSC trap: Trace who pays now, later, through taxes or poor service.

Q11. Which description of open access is most defensible?

A. It provides free use of electricity networks. B. It abolishes universal-service obligations. C. It permits eligible users to choose supply while paying applicable network and system charges. D. It guarantees every purchase is renewable.

Answer: C

Option explanations:

- **A is incorrect:** Network use carries wheeling, transmission, loss and other lawful charges.
- **B is incorrect:** Distribution obligations continue under statute and licence.
- **C is correct:** Choice is combined with cost recovery for monopoly wires and system services.
- **D is incorrect:** The source depends on the transaction; green open access is a specified subset.

UPSC trap: Supplier choice does not mean costless wires.

Q12. Which statement about power exchanges is correct?

A. Day-ahead price is the retail tariff for all consumers. B. Exchanges replace every long-term PPA. C. Market coupling was universally complete before July 2025. D. They provide regulated short-term products while much procurement remains outside exchanges.

Answer: D

Option explanations:

- **A is incorrect:** Exchange price excludes many retail network, loss and policy components.
- **B is incorrect:** Long-term and bilateral contracts continue.
- **C is incorrect:** CERC's July 2025 direction initiated phased work rather than proving completion.
- **D is correct:** Day-ahead, real-time and term-ahead products form part of the procurement architecture.

UPSC trap: Do not generalise one short-term price to the whole sector.

Q13. What is the proper role of deviation settlement?

A. It prices departures from schedules to support discipline and balance. B. It is a long-term capacity contract. C. It replaces real-time system operation. D. It is a household efficiency subsidy.

Answer: A

Option explanations:

- **A is correct:** DSM compares actual injection or drawal with schedule and applies settlement.
- **B is incorrect:** Capacity contracts procure dependable capability over another horizon.
- **C is incorrect:** Operators still deploy reserves and manage congestion.
- **D is incorrect:** Demand-side efficiency is unrelated despite sharing the abbreviation DSM.

UPSC trap: Spell out DSM because it has two common meanings.

Q14. Which statement about ancillary services is correct?

A. They are measured only by annual MWh. B. They procure response and reserves needed for frequency and reliability. C. They are identical to cross-subsidy. D. Only coal plants can provide them.

Answer: B

Option explanations:

- **A is incorrect:** Response speed and availability matter in addition to energy.
- **B is correct:** Ancillary products support balancing and contingencies.
- **C is incorrect:** Cross-subsidy concerns tariff incidence between classes.
- **D is incorrect:** Hydro, storage, gas, demand response and eligible generators can contribute.

UPSC trap: Energy and reliability services are distinct products.

Q15. AT&C loss primarily combines:

A. Only transformer heat loss. B. Only unmetered farm supply. C. Technical, billing and collection losses. D. The difference between coal and gas prices.

Answer: C

Option explanations:

- **A is incorrect:** Physical loss is only one component.
- **B is incorrect:** Unmetered supply may affect estimation but does not exhaust the metric.
- **C is correct:** The metric links input energy to revenue actually collected.
- **D is incorrect:** Fuel-price comparison belongs to generation economics.

UPSC trap: AT&C is broader than theft and narrower than total financial loss.

Q16. Which Ministry of Power FY 2024-25 pair is correct?

A. AT&C 21.91%; ACS-ARR Rs 0.69/kWh B. AT&C 12%; ACS-ARR zero C. AT&C 5.28%; ACS-ARR Rs 15.04/kWh D. AT&C 15.04%; ACS-ARR Rs 0.06/kWh

Answer: D

Option explanations:

- **A is incorrect:** These are the FY 2020-21 starting values in the comparison.
- **B is incorrect:** These are RDSS targets, not the reported outcome pair.
- **C is incorrect:** The numbers and units are reversed; 5.28 crore refers to meter installations.
- **D is correct:** The annual report gives this FY 2024-25 national pair.

UPSC trap: Keep percentage loss and rupees-per-kWh gap in their units.

Q17. How should UDAY and RDSS be distinguished?

A. UDAY centred on a 2015 debt reset plus reforms; RDSS is a later conditional infrastructure and performance scheme. B. RDSS is a new name for UDAY debt bonds. C. UDAY began after RDSS. D. Both are sections of the Electricity Act.

Answer: A

Option explanations:

- **A is correct:** The schemes differ in date, instrument and principal mechanism.
- **B is incorrect:** RDSS works and smart metering are not the old debt takeover.
- **C is incorrect:** UDAY launched in November 2015; RDSS followed later.
- **D is incorrect:** They are policy schemes, not sections of the Act.

UPSC trap: A successor reform is not automatically a merger.

Q18. What does a zero ACS-ARR gap fail to prove by itself?

A. That a tariff order exists. B. That subsidy, dues and collections arrived on time and service improved. C. That average cost can be measured. D. That power purchase has a cost.

Answer: B

Option explanations:

- **A is incorrect:** A tariff order can exist regardless of the gap.
- **B is correct:** Accrual accounting can conceal cash timing, arrears and quality problems.
- **C is incorrect:** The indicator presupposes measured cost and revenue.
- **D is incorrect:** Power purchase remains a major ACS component.

UPSC trap: Pair booked gap with cash flow and service quality.

Q19. Which coal statement is correct?

A. Coking and non-coking coal have identical uses. B. Imports prove India has no domestic production. C. Production, dispatch, plant receipt and generation are distinct stages. D. A mine award guarantees immediate output.

Answer: C

Option explanations:

- **A is incorrect:** Coking coal is crucial for metallurgical use; non-coking serves power and others.
- **B is incorrect:** India produced 1,047.523 MT in FY 2024-25 while importing coal.
- **C is correct:** Quality and logistics create gaps between each stage.
- **D is incorrect:** Clearances, development and evacuation precede production.

UPSC trap: Follow the physical chain instead of treating one quantity as another.

Q20. Which pair accurately states FY 2025-26 coal imports?

A. Total 66.33 MT, all coking B. Total 180.04 MT, all non-coking C. Total 1,047.523 MT, all imports D. Total 246.37 MT: 66.33 coking and 180.04 non-coking

Answer: D

Option explanations:

- **A is incorrect:** 66.33 MT is only the coking component.
- **B is incorrect:** 180.04 MT is only the non-coking component.
- **C is incorrect:** 1,047.523 MT is FY 2024-25 domestic production.
- **D is correct:** The two components sum to the Ministry's reported total.

UPSC trap: Production and import flows require the right year and category.

Q21. Which statement about revised SHAKTI is accurate?

A. It is a power-sector coal-linkage framework revised with Cabinet approval on 7 May 2025. B. It regulates city-gas pipelines. C. It is a renewable-certificate market. D. It abolishes every coal auction.

Answer: A

Option explanations:

- **A is correct:** SHAKTI governs specified coal-allocation and linkage routes for power.
- **B is incorrect:** PNGRB regulates specified gas-network activities.
- **C is incorrect:** REC operates under CERC regulations.
- **D is incorrect:** The reform does not erase every auction or contract.

UPSC trap: Coal linkage, mine auction and e-auction are different.

Q22. Why can a coal plant become economically stranded?

A. Every plant must close at the same age. B. Lower utilisation, policy change or cheaper alternatives can erode cash flow before debt recovery. C. Sunk cost is always recoverable. D. Environmental controls have no cost.

Answer: B

Option explanations:

- **A is incorrect:** Retirement depends on economics, policy and system need.
- **B is correct:** Unexpected utilisation and revenue changes can destroy asset value.
- **C is incorrect:** Recovery depends on contracts and regulatory prudence.
- **D is incorrect:** Retrofits can require substantial capital and operating spending.

UPSC trap: Low PLF alone does not prove stranding.

Q23. Which petroleum-chain mapping is correct?

A. Upstream refining; downstream exploration B. Midstream retail pricing; upstream city gas C. Upstream exploration/production; midstream transport/storage; downstream refining/marketing D. PNGRB regulates only upstream production

Answer: C

Option explanations:

- **A is incorrect:** Refining is downstream and exploration is upstream.
- **B is incorrect:** Retail is downstream; city gas is a regulated network activity.
- **C is correct:** This is the standard functional chain.
- **D is incorrect:** PNGRB expressly excludes crude and gas production.

UPSC trap: Classify the activity before naming its regulator.

Q24. Which PPAC FY 2025-26 provisional inference is correct?

A. India produced more crude than products consumed. B. Gas imports were zero. C. Crude and gas had identical dependence. D. Crude dependence was 88.7%, while gas dependence was 50.1%.

Answer: D

Option explanations:

- **A is incorrect:** Crude output was 28.0 MMT and product consumption 241.6 MMT, although categories differ.
- **B is incorrect:** LNG imports were 34,427 MMSCM.
- **C is incorrect:** The ratios differ and use commodity-specific denominators.
- **D is correct:** This reproduces PPAC's ratios with the provisional-year caveat.

UPSC trap: Do not infer refinery weakness from crude dependence.

Q25. What is PNGRB's statutory perimeter?

A. Specified downstream/network petroleum and gas activities, excluding crude and gas production. B. All electricity tariffs. C. Coal-mine safety. D. Nuclear fuel regulation.

Answer: A

Option explanations:

- **A is correct:** This is the perimeter stated by PNGRB under its 2006 Act.
- **B is incorrect:** Electricity commissions regulate electricity tariffs.
- **C is incorrect:** Coal institutions perform mining functions.
- **D is incorrect:** Atomic-energy institutions govern nuclear fuel.

UPSC trap: The upstream-production exclusion is a frequent trap.

Q26. What does a strategic petroleum reserve do?

A. Permanently eliminates imports. B. Provides a finite physical buffer while wider diversification remains necessary. C. Hedges only the exchange rate. D. Guarantees indefinite retail-price stability.

Answer: B

Option explanations:

- **A is incorrect:** Finite inventory cannot replace continuing supply.
- **B is correct:** Stocks buy response time and complement source, route and contract diversity.
- **C is incorrect:** Financial instruments hedge price or currency; a reserve supplies crude.
- **D is incorrect:** Drawdown is limited and cannot neutralise every long shock.

UPSC trap: Capacity, fill level and days of cover are separate.

Q27. Which statement about renewable auctions is correct?

A. Tendered capacity is already generating. B. The winning bid is the complete retail tariff. C. A low bid still depends on finance, land, grid, commissioning and buyer payment. D. Auctions eliminate curtailment.

Answer: C

Option explanations:

- **A is incorrect:** Tender and commissioning are different stages.
- **B is incorrect:** Network, balancing and retail components remain.
- **C is correct:** These delivery conditions determine whether a bid becomes service.
- **D is incorrect:** Congestion and security constraints can still curtail output.

UPSC trap: Follow status from announcement to generation.

Q28. Which statement best explains curtailment?

A. It is always lack of sun or wind. B. It is storage discharge. C. It proves plant inefficiency. D. Available generation may be reduced because of congestion, security or demand constraints.

Answer: D

Option explanations:

- **A is incorrect:** Weather unavailability is not curtailment of available energy.
- **B is incorrect:** Storage discharge adds supply rather than withholding it.
- **C is incorrect:** A plant may be technically available when the grid cannot accept output.
- **D is correct:** This definition preserves the system-operator boundary.

UPSC trap: Variability, outage and curtailment are not synonyms.

Q29. Which GEC-II statement is correct?

A. It targets about 10,750 ckm and 27,500 MVA to integrate about 20 GW in seven states. B. It is a coal-mine scheme. C. Its design proves all lines are complete. D. It finances only rooftop panels.

Answer: A

Option explanations:

- **A is correct:** These are MNRE's official scheme-design figures.
- **B is incorrect:** The corridor concerns renewable-power transmission.
- **C is incorrect:** Targets do not establish completion.
- **D is incorrect:** The scheme builds intra-state transmission infrastructure.

UPSC trap: Circuit-km, MVA and GW measure different things.

Q30. Which statement about storage is correct?

A. Storage generates primary energy. B. Storage adds timing and reliability value but returns less energy than it absorbs. C. A MW rating fully states duration. D. Pumped storage has no ecological constraints.

Answer: B

Option explanations:

- **A is incorrect:** Storage shifts converted electricity; it does not create primary energy.
- **B is correct:** Round-trip loss is exchanged for timing and system service.
- **C is incorrect:** MWh and the MW/MWh ratio are required.
- **D is incorrect:** PSP depends on site, water, ecology and civil works.

UPSC trap: No storage claim is complete without power, energy and efficiency.

Q31. Which green-hydrogen status statement is accurate?

A. The 2030 target was achieved in 2023. B. Electrolyser capacity equals hydrogen output. C. The target and awards must be separated from commissioned production. D. Green hydrogen is mined.

Answer: C

Option explanations:

- **A is incorrect:** The mission sets a future annual-production target.
- **B is incorrect:** Electrolyser and hydrogen capacity use different units.
- **C is correct:** The Survey reports allocation and awards, not automatic operating output.
- **D is incorrect:** Hydrogen is produced as an energy carrier.

UPSC trap: Target, allocation, award and commissioning are separate.

Q32. Which comparison is correct?

A. REC and carbon credit are always the same. B. PAT ESCert certifies renewable generation. C. ESO is a crude-stock obligation. D. REC represents eligible renewable MWh; carbon credits and ESCerts have different bases.

Answer: D

Option explanations:

- **A is incorrect:** Renewable attributes and GHG units have different rules.
- **B is incorrect:** ESCert represents verified energy saving.
- **C is incorrect:** ESO concerns energy storage in the notified trajectory.
- **D is correct:** The instruments are not automatically fungible.

UPSC trap: One REC equals one eligible MWh, not one tonne CO2e.

PYQS AND ANSWER PRACTICE

Official-key discipline: Mains wording below is reproduced from the verified local official-paper route. These are learner model answers, not UPSC model answers. An objective answer letter is shown only after an exact authenticated question-paper/set-to-key mapping; no letter is inferred from a neutral route.

Verified Mains PYQ 1 - GS-III 2018 - 10 marks, 150 words

Question: "Access to affordable, reliable, sustainable and modern energy is the sine qua non to achieve Sustainable Development Goals (SDGs)." Comment on the progress made in India in this regard.

Model answer: India has moved from access scarcity toward service-quality and transition challenges. Electrification expanded network reach, while the Ministry of Power reports FY 2024-25 national energy and peak deficits of only 0.1% and 0.0%. Reliable electricity supports health facilities, digital education, irrigation and enterprise; clean cooking reduces indoor pollution and women's time burden. Yet connection is not continuous supply: affordability, voltage, outages, refill cost and productive use remain uneven. DISCOM weakness matters because a financially stressed last mile cannot maintain wires or procure power. Sustainability also requires efficiency, renewable integration, storage and cleaner fuels rather than only more capacity. Thus progress is substantial in physical access and aggregate adequacy, but SDG 7 demands an outcome test: affordable hours of quality power, clean cooking actually used, and resilient low-emission service for vulnerable households.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Verified Mains PYQ 2 - GS-III 2020 - 15 marks, 250 words

Question: Describe the benefits of deriving electric energy from sunlight in contrast to the conventional energy generation. What are the initiatives offered by our Government for this purpose?

Model answer: Solar electricity has no fuel purchase or combustion at generation, is modular and quick to deploy, and can serve utility, rooftop, pump and remote loads. It reduces marginal fuel-import exposure and local air pollution compared with fossil generation. Government instruments include solar parks and competitive procurement; PM Surya Ghar for rooftops; PM-KUSUM for farm applications; Green Energy Corridors; renewable obligations and RECs; domestic-manufacturing support; and storage procurement/VGF. Economic Survey 2025-26 reports 135.81 GW installed solar capacity by December 2025 and 8 GW rooftop capacity under PM Surya Ghar, but installed MW is not annual generation. Solar output is daytime-variable, needs land and transmission, and creates evening ramps, mineral and recycling issues. Solar's advantage is therefore strongest when capacity is integrated with grids, storage, demand response, domestic capability and ecological siting, and judged by reliable delivered energy rather than the winning bid.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Verified Mains PYQ 3 - GS-III 2021 - 10 marks, 150 words

Question: Explain the purpose of the Green Grid Initiative launched at World Leaders Summit of the COP26 UN Climate Change Conference in Glasgow in November 2021. When was this idea first floated in the International Solar Alliance?

Model answer: The Green Grids Initiative-One Sun One World One Grid seeks interconnected regional and global grids that move renewable electricity across time zones. A wider balancing area can use geographic diversity, share reserves, reduce curtailment and improve access to clean power. India first presented the transnational solar-grid idea at the International Solar Alliance's First Assembly in October 2018. Its economic logic is that sunlight available in one region can serve evening demand elsewhere through transmission. Yet cross-border lines do not automatically create trade. Common technical codes, transparent scheduling and settlement, finance, cyber security, sovereignty safeguards and mutual trust are required. It complements rather than replaces domestic corridors, storage and demand response. Interconnection can lower balancing cost and improve resilience only when market rules and political risks are managed.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Verified Mains PYQ 4 - GS-III 2022 - 15 marks, 250 words

Question: Do you think India will meet 50 percent of its energy needs from renewable energy by 2030? Justify your answer. How will the shift of subsidies from fossil fuels to renewables help achieve the above objective? Explain.

Model answer: The target must be stated correctly: India's COP26 commitment concerned about 50% of cumulative installed electric-power capacity from non-fossil sources by 2030, not half of total primary-energy consumption. India had reached 51.93% non-fossil installed capacity by 31 December 2025, while non-fossil generation was 30.41% in April-December 2025. Capacity therefore does not settle reliable energy contribution. Shifting support can improve relative prices by reducing general fossil consumption or producer support, financing renewable auctions, grids, storage, efficiency and targeted access, and internalising pollution and import risk. Abrupt withdrawal of lifeline electricity or clean-cooking support would be regressive; coal regions, workers and legacy debt need transition assistance. Renewable subsidies can also cause land, manufacturing or fiscal distortions. The appropriate shift is from general price suppression and unpriced externalities toward transparent, competitive support for generation, transmission, storage, demand flexibility and vulnerable users. Success should be measured by generation, curtailment, system cost, reliability and emissions, not capacity alone.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Verified Mains PYQ 5 - GS-III 2025 - 10 marks, 150 words

Question: How can India achieve energy independence through clean technology by 2047? How can biotechnology play a crucial role in this endeavour?

Model answer: Energy independence should mean lower strategic vulnerability, not autarky. India can reduce exposure through efficiency, electrification, diversified renewable and nuclear supply, stronger grids, storage, green hydrogen for hard-to-abate uses, domestic clean-technology manufacturing, critical-mineral partnerships and recycling. PPAC's provisional FY 2025-26 crude dependence of 88.7% shows why transport and industrial substitution matter, while reliability still requires firm capacity and flexible demand. Biotechnology can produce advanced biofuels, biogas and sustainable aviation-fuel feedstocks from residues, improve energy crops and microbial conversion, and support waste-to-energy pathways. Food-water-land competition, lifecycle emissions, biodiversity, logistics and cost must be audited. A resilient 2047 pathway combines demand reduction, clean carriers, domestic capability and diversified imports, judged by delivered cost, emissions and security rather than literal zero imports.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Objective PYQ official-key discipline

Year	Routed demand	Key treatment
2018	Solar power, silicon wafers and tariff regulation	Answer withheld: no authenticated local final key mapping used.
2019	PNGRB role and appellate route	Answer withheld: no authenticated local final key mapping used.
2022	Coal Controller's Organisation functions	Answer withheld: no authenticated local final key mapping used.
2023	Coal thermal plants, seawater and water stress	Answer withheld: no authenticated local final key mapping used.
2025	PNGRB regulated activities	Official Set-A key exists locally, but the full official option set is not reproduced here; no letter is inferred.
2025	Brazil/USA ethanol feedstock comparison	Official Set-A key exists locally, but the full official option set is not reproduced here; no letter is inferred.
2026	Green-hydrogen pathways and mission	Repository route labels the key provisional; no answer letter is recorded.

Rule: Concept teaching never manufactures an official option from a neutral routing summary.

ORIGINAL MAINS 1 - 10 MARKS - 150 words

Question: Distinguish installed capacity, generation, peak demand, energy supplied and final consumption. Why does the distinction matter for policy?

Model answer (138 native-body words):

Installed capacity is the rated power of generating assets, measured in MW or GW. Generation is electrical energy actually produced during a period, measured in MWh, GWh or BU. Peak demand is the highest instantaneous requirement; peak met records the maximum supplied at that interval. Energy supplied is the cumulative electricity delivered during a period. Final consumption is energy used by end-users after network and transformation losses and may include fuels beyond electricity. These distinctions change policy conclusions. At 31 December 2025, non-fossil sources were 51.93% of installed power capacity, but their April-December 2025 generation share was 30.41%. Storage also needs both MW and MWh. A credible evaluation must therefore attach unit, period, denominator and system boundary to every figure, then test whether capacity is available at peak, reaches consumers reliably and produces affordable useful service.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 2 - 10 MARKS - 150 words

Question: Why are DISCOM finances central to reliable electricity supply? Suggest a compact reform sequence.

Model answer (128 native-body words):

DISCOMs are the commercial hinge between generators and consumers. Inflow depends on accurate metering, billing, collection, timely state subsidy and government-dues payment; outflow includes power purchase, maintenance, employees, finance and capital works. Technical loss, theft, weak billing, delayed tariffs, unpaid subsidy and inflexible PPAs can create a cycle of borrowing, generator dues and poor service. Ministry of Power reports FY 2024-25 AT&C loss of 15.04% and ACS-ARR gap of Rs 0.06/kWh, but national averages do not prove every utility is healthy. Reform should sequence feeder and transformer accounting, reliable meters and grievance systems, targeted lifeline support, timely tariffs and subsidy payment, procurement review, loss-reduction investment and transparent regulation. RDSS should be judged by validated cash recovery, outage reduction and service, not meter installation alone.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 3 - 15 MARKS - 250 words

Question: Analyse the economics of integrating variable renewable energy into India's power system.

Model answer (175 native-body words):

Variable renewable energy has low fuel and short-run marginal cost, but output depends on weather and time. The problem shifts from procuring cheap plant-level energy to delivering reliable system service. At 31 December 2025, non-fossil sources formed 51.93% of installed capacity, while their April-December 2025 generation share was 30.41%; the different periods notwithstanding, this shows why nameplate capacity is not firm supply. Integration requires advance transmission, accurate forecasting, larger balancing areas, flexible hydro and thermal operation, storage, demand response, reserves and short-term markets. MNRE's GEC-II design aims at 10,750 circuit-km and 27,500 MVA to integrate about 20 GW. Economic Survey 2025-26 cites storage need of 336 GWh by 2029-30 and 411 GWh by 2031-32. Costs include curtailment, networks, balancing, land, minerals and recycling; benefits include avoided fuel, pollution and imports. Policy should procure portfolios and services rather than isolated megawatts: generation plus evacuation, duration-specific storage, time-of-day demand and resource adequacy. The verdict is pro-integration: success is lower unserved energy and total system cost, not the lowest bid or largest capacity addition.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 4 - 15 MARKS - 250 words

Question: Evaluate coal's changing role in India's energy security and transition.

Model answer (174 native-body words):

Coal remains central because existing plants, domestic mines and rail-linked infrastructure supply large volumes of dispatchable electricity and industrial fuel. The Ministry of Coal reports 1,047.523 MT production in FY 2024-25, yet FY 2025-26 imports were 246.37 MT, including 66.33 MT coking coal. Security depends on grade and delivered logistics, not aggregate production alone. Revised SHAKTI, approved 7 May 2025, addresses power-sector linkages, while commercial mine auctions concern another allocation route. Coal also imposes air, water, ash, land and carbon costs. As solar and wind expand, thermal plants may run fewer hours but need better ramping and reserve performance; lower PLF can raise per-unit fixed-cost recovery. Abrupt retirement can strand PPAs and debt and concentrate job, royalty and local-economy losses. Policy should prioritise efficient units, logistics, pollution compliance, flexible operation, transparent retirement criteria and closure funds while scaling grids, storage and demand response. A just transition needs worker protection, land restoration and regional diversification. Coal's role should decline through planned substitution of energy and reliability services, not a capacity-only pledge.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 5 - 20 MARKS - 250 words

Question: Assess India's oil and gas security architecture in the face of geopolitical and price shocks.

Model answer (186 native-body words):

India's vulnerability begins with high import dependence but extends to supplier concentration, shipping routes, currency, contracts, refining configuration and pricing. PPAC's FY 2025-26 provisional data report crude dependence of 88.7% and gas dependence of 50.1%. High crude dependence can coexist with strong refining and product exports, so the denominator matters. A shock raises the import bill, pressures the current account and rupee, transmits through fuel, freight and inputs, and creates a fiscal-monetary trade-off. The response requires layers. Diverse suppliers and routes reduce concentration; long contracts secure volume while spot cargoes retain flexibility; refinery complexity broadens crude choice; LNG terminals and common-carrier pipelines diversify gas access; efficiency and electrification reduce demand. The 5.33 MMT Phase-I strategic reserve is a finite buffer, while approved 6.5 MMT Phase-II capacity must not be counted as operating stock. Financial hedges manage price but not physical blockage. Targeted transfers are preferable to prolonged general price suppression. PNGRB's pipeline access can deepen markets, but affordability and anchor demand remain constraints. Security is diversified interdependence with buffers and substitution, measured by disruption recovery and delivered cost rather than zero imports.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 6 - 20 MARKS - 250 words

Question: Design a policy framework for a reliable, affordable and just Indian energy transition.

Model answer (189 native-body words):

A just transition must coordinate technology, networks, finance and distribution. First, plan from demand and reliability: distinguish MW from MWh, forecast peak and energy requirements, and procure dependable portfolios. Second, integrate renewables through corridors, storage, flexible hydro and thermal units, demand response, ancillary services and adequacy rules. Third, repair distribution through energy accounting, meters, timely tariffs and subsidy, targeted lifeline support and disciplined PPAs; FY 2024-25 national AT&C loss of 15.04% and ACS-ARR gap of Rs 0.06/kWh are progress indicators, not proof of universal health. Fourth, keep RPO/RCO, ESO, REC, PAT ESCerts and carbon credits distinct and prevent double counting. Fifth, reduce external vulnerability through efficiency, electrification, diversified oil and gas contracts, strategic stocks, domestic clean-tech capability, critical-mineral partnerships and recycling. Sixth, price pollution while financing reskilling, mine closure, regional diversification and clean cooking. VGF should buy measurable system services and not socialise avoidable project risk. Finally, publish a dashboard of unserved energy, outage quality, system cost, fiscal exposure, emissions, import concentration and distributional incidence. Success means dependable low-carbon service without shifting hidden debt, ecological damage or adjustment costs onto vulnerable consumers and regions.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.